

Vascularization reconstruction using Contrast-Enhanced Micro CT for high-resolution X-ray based 3D histology

Nachtrab Sean

↳ adapt later → you need
to add more on what
your thesis will do
→ this is what CECT can do

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Master of Science: SINF - Computer science with elective Medical informatics

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Abstract

The accurate reconstruction of tissue vascularization in ex-vivo biopsies is essential to quantitative analysis of samples, to enable their comparison across various treatment profiles, and achieve the level of accuracy required for drug research. We aim to enable the analysis of antiangiogenic drugs for cancer treatment, specifically focused on pazopanib, by improving quantification of vascularization. Classical histology methods such as 2D histology suffer from a lack of accuracy due to the slicing process, and when used for vascularization reconstruction, lose the ability to reconstruct its structural nature. 3D techniques such as contrast enhanced Micro CT enables the acquisition of high resolution 3D images of ex-vivo samples aided by a contrast agent, but does not enable direct quantitative information extraction due to discontinuities, noise, low contrast and small features. In this document, a method for the structural reconstruction of vascularization from microCT images is presented, enabling analysis of the impact of antiangiogenic drugs on the vascularization of tumors in mice.

-> this can be improved and extended later

to do so
after step 1
we can
→ find better
descriptions

Check exact definition of histology

Chapter 1

Introduction (2-6pp) (does this overlap with state-of-the-art and should I remove the intro?)

The analysis of soft tissue is foundational to modern medicine. It is used everywhere from clinical environments to laboratories, where samples are collected and imaged in order to obtain an understanding of the tissue, called histology or histopathology in the case of diseased tissue. We use the term histology to refer to either interchangeably. Classically, this sampling and subsequent analysis has been done in 2D, with the microscope being the catalyst for the development of the field, and being the gold standard for this process. However, this form of histology presents multiple limitations that stem from its 2D nature: the loss of 3D understanding and structure, as well as very limited quantification capabilities. [0.1] [More detail to be added]

Histology is relevant in both clinical and laboratory environments. Clinically, the goal of a biopsy and subsequent analysis under the microscope is generally to identify a qualitative parameter in order to enable diagnosis and treatment: identifying the presence of a kind of cell, the phenotyping of a cell, or the estimation of a parameter, for example in cancerology the presence or absence of blood vessels in a tumor. [0.2]

Such biopsies done in 2D present the distinct disadvantage of being unable to obtain information at the same granularity in all three axis: microscope slides in 2D may be stacked to construct a 3D understanding, but these stacked planes are separated by a dead space where no information is acquired [0.3]. This loss of accuracy is catastrophic for the reconstruction and quantification of structures that are small and where their structure is relevant to their function, such as blood vessels, on top of the deformation caused by the cutting process.

Blood vessel understanding and reconstruction has emerged over the past years as a critical element in understanding and pushing the boundaries in new drug research: angiogenesis, or the creation of new blood vessels, is a key step in the growth and spread of cancerous tumors. Classical methods, relevant for qualitative understanding and estimation, fall short in terms of accuracy for quantitative analysis for clinical trials [0.4]. The 2D reconstruction:

check the exact definition of "histology"

1-2 pages

~~50%~~

about 15-20 pages

don't 2D histology = 2D!

is it really about research? not only clinical

is also ones in 1 sheet for

enough others, yes

of that?

yes!
+ clinical

1. (DOES THIS OVERLAP WITH STATE-OF-THE-ART AND SHOULD I REMOVE THE INTRO?)

its variability and lack of resolution being a limiting factor, the field is progressing towards the use of high resolution 3D data for these quantitative analysis tasks.

We will begin by diving into the current state of the art in histology for drug screening, followed by the MicroCT methods used, and finish with an exploration of the methods of image understanding more broadly,

Should this be extended with the aetiology of cancer and angiogenesis?

0.1 Appears interesting: Limitations of clinical and biological histology <https://doi.org/10.1054/mehy.1999.0894>

0.2 They speak of "presence of vascular or stromal invasion": Histology: The gold standard for diagnosis? <https://pubmed.ncbi.nlm.nih.gov/37008634/>

0.3 a. 3D reconstruction from 2D slices "Extending two-dimensional histology into the third dimension through conventional micro computed tomography" <https://doi.org/10.1016/j.neuroimage.2016.06.005>

0.3 b. Other source for 3D reconstruction from 2D slices "A Survey of Methods for 3D Histology Reconstruction" <https://doi.org/10.1016/j.media.2018.02.004>

0.4 Sources on angiogenesis in tumors from Wlodarski

you should come with
CECT being a new tool
for 3D histology!

→ BUT limitations = reconstruction
of the vasculature
network in 3D

please check my
lips 2 slides
slides

you should come with CECT being a new tool for
3d biology but limitations = reconstruction of the
vasculature network in 3d -> here is where the SOTA
will continue on + thesis topic will work on

here is where the SOTA will
continue on

+ your thesis topic
will work on

Chapter 2

State of the art

State of the art from "Scientific writing-tips and tricks_Deadlines.pdf":

In the state of the art, you indicate what has been studied, why it has been studied, and, in general terms, how it has been studied.

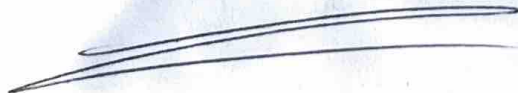
This section contains:

- ✓ The background of the study (= context);
- ✓ Research that has been done in the frame of your thesis topic;
- ✓ Information that is needed for the reader to understand your topic and the remaining scientific issues/problems.

→ Make a fluent story of it, and not just a sum up of different papers or references.

→ The context (per paragraph or section) usually ends with a conclusion or problem statement that sets the scene for your specific research questions and project aim + objectives.

yes :-)



2. STATE OF THE ART

first give the definition of histology

With the goal of reconstructing tumor vascularization for the analysis of the antiangiogenic drug Pazopanib in the context of cancer treatment, we begin by exploring the use of histology for drug screening, followed by imaging methods, more specifically contrast-enhanced Micro CT, and finish with an exploration of the methods of structure extraction from images.

Classical 2D

SOTA is like a funnel

2.1 Histology for drug screening

In order to bring new drugs to market, a set of steps must be followed that are established by organisations such as the European Medical Association in the European Union, or the Food and Drug Administration in the U.S [1.1]. One of the key steps is collecting data on the effectiveness of the drug, done through clinical trials. These trials, before taking place in humans, are done on animals in a controlled lab environment, where collected data is analyzed for the outcomes and side effects of the drug. In the analysis process for Pazopanib, tumors are sampled and analysed ex-vivo, the histology of these tumors is critical in identifying the impact of treatment when compared to a control.

The conclusions taken from each step in the clinical trial decide if the trial moves on, meaning that a more accurate prediction with tighter confidence intervals enables a better discriminative power and reduces the amount of effective drugs discarded due to uncertainty from process limitations.

2.1.1 Histology in industry

Histology is in use in industry for clinical trials amongst other tasks, and the field is growing [2.1]. Histology is used due to its large amounts of qualitative data and high discriminative power, and is interpreted by experts during the process of drug research. It is used in clinical trials of structure altering drugs, or where the effect of interest is expected to be visible, and is generally done by specialized histopathologists [2.1]. Clinical trials vary widely in their design, as well as have a large variety of possible targets, in oncology specifically, 2D histology remains the most widely used technique [2.3], [2.27, 2.60] for the histology of human biopsies. For the analysis of Pazopanib, 2D histology is the reference method used to evaluate tumors.

Classical 2D histology

The field of histology is born from 2D histology of tissue under microscope magnification. 2D microscope slice histology, known as "classical histology", takes the form of tissue sample collection, often embedding of the tissue, followed by staining with various agents, applied to increase contrast or highlight certain structures, and ending in investigation using optical or electron microscopy [3.1]. This technique is used due to its delivery of a large amount of relevant quantitative data immediately interpretable by experts, and high discriminative power, although it is time consuming and is costly as a result [3.2]. As a technique it has evolved in multiple directions, from the utilization of advanced staining agents [3.2] allowing the staining of a broad range of classes (DNA, proteins, lipids, or carbohydrates), to different lighting wavelengths, ranging from ultraviolet [3.3] to infrared [3.4].

2D histology is limited by its requirement to slice the target tissue before analysis, resulting in destructive modification of the sample. This slicing has the potential of deforming the

• Tumor angiogenesis

• Drug screening for tumor angiogenesis

• Classical 2D histology

• Key based 3D histology

• 3D reconstruction of using "classical" registration

• Deep learning for vessel registration

and the the current problem is lack of generalization of your results + maybe defective